Stochastic Models and Applications

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Reference Material

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- V.K. Rohatgi and A.K.Md.E. Saleh, An Introduction to probability and Statistics, Wiley, 2nd edition, 2018
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- Scott Sheffield, Probability and Random Variables, Massachusetts Institute of Technology, MIT OpenCourseWare: https://ocw.mit.edu/courses/mathematics/18-600-probability-and-random-variables-fall-2019/
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  - continuity, differentiability, derivatives, functions of several variables, partial derivatives, integration, multiple integrals or integration over $\Re^n$, convergence of sequences and series, Taylor series

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  But we would review these basic probability concepts.

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- It is also useful in many engineering systems, e.g., for analyzing behaviour under noise.

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This is only a 'sample' of possible application scenarios!

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- Notation: $\phi \in 2^\Omega$ denotes the null set.

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- How does one assign probabilities for these events?

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$(\Omega, \mathcal{F}, P)$ is called the **Probability Space**

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- As defined, the co-domain of the function P is $\Re$. However, the axioms imply that it takes values in $[0, 1]$.

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$$P(B) = P(A + (B - A)) = P(A) + P(B - A) \geq P(A)$$

  Thus, $A \subset B \Rightarrow P(A) \leq P(B)$

  Also, $P(B - A) = P(B) - P(A)$ when $A \subset B$.

- If A is an event, then $A \subset \Omega$ and hence $P(A) \leq P(\Omega) = 1$.

- We can show $P(A^c) = 1 - P(A)$ as

$$1 = P(\Omega) = P(A + A^c) = P(A) + P(A^c)$$

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$$P(U_{i=1}^{n} A_i) = \sum_i P(A_i) - \sum_i \sum_{j>i} P(A_i \cap A_j)$$
$$+ \sum_i \sum_{j>i} \sum_{k>j} P(A_i \cap A_j \cap A_k) - \cdots + (-1)^{n+1} P(\cap_i A_i)$$

Known as inclusion-exclusion formula

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- This is the usual familiar formula: number of favourable outcomes by total number of outcomes.
- Thus, 'equally likely' is one way of specifying the probability function (in case of finite Ω).

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- Using 'equally likely' is how we are 'modeling' this situation.

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- We can use this idea even when Ω is countably infinite

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- This is how we normally define a probability measure on countably infinite Ω.

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- Consider a random experiment of tossing a biased coin repeatedly till we get a head. We take the outcome of the experiment to be the number of tails we had before the first head.
- A (reasonable) probability assignment is:

$$P(\{k\}) = (1-p)^k p, k = 0, 1, \cdots$$

where p is the probability of head and $0 < p < 1$.
(We assume you understand the idea of 'independent' tosses here).

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(There are many issues that need more attention here).

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- We can take $\Omega = \{(x, y) \ : \ 0 < x < y < 1\} \subset \Re^2$.
- For the pieces to make a triangle, sum of lengths of any two should be more than the third.

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- So the event of interest is:

$$A = \{(x,y) \in \Omega \;:\; y > 0.5;\; x < 0.5;\; y < x + 0.5,\}$$

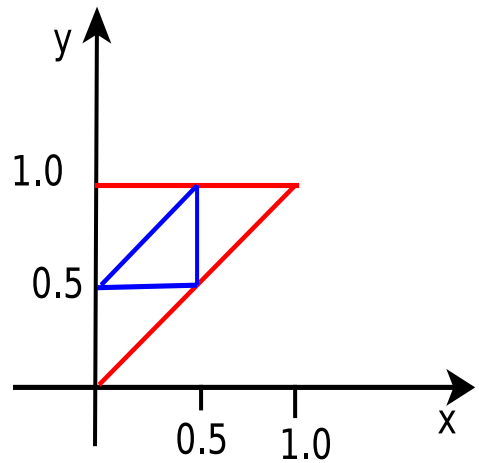
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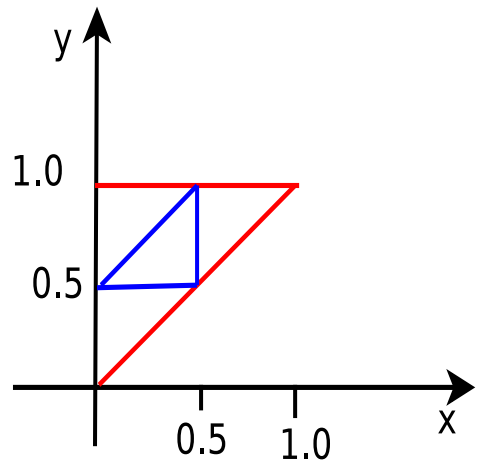
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- The required probability is area of A divided by area of Ω which gives the answer as 0.25

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- Once we understand condional probability is a new probability assignment, we go back to the 'standard notation'

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- It is not as if we talk about conditional probability only for subsets of B. Conditional probability is also with respect to the original probability space. Every element of $\mathcal{F}$ has conditional probability defined.

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- There are probabilistic methods to capture causation (but far beyond the scope of this course!)

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- This is very useful in many situations. ("arguing by cases")